



RoboDesk for wheelchair-bound people (Image courtesy: wfdd.org)

speech into text for two-way communication. UNI also allows users to create their own sign language with its sign builder, so it is easy to add custom language to dictionaries. It is a subscription-based app with two versions, one requires a data connection while the other doesn't.

Transcence. Transcence is a mobile app that makes group conversations possible for hearing-impaired people. In a conversation, using the microphone of each participant's smartphone, the app catches what the participant is saying, then converts it into text in real time. All speakers are assigned corresponding text bubbles, differentiated by colour, just like what we would find in a regular group messaging chat room.

Devices for people with speech impairment

DynaVox EyeMax. This device gives paralysis, cerebral palsy and stroke victims the ability to participate in spoken communication using only their eyes. Using a sophisticated eye tracking system, users can interact with an on-screen keyboard to enter words and phrases. These words and phrases are then translated into

spoken text using text-to-speech mechanism of the device. In addition to the on-screen keyboard, the language software on the device offers hundreds of pre-defined phrases and words, which can be selected from lists or chosen via pictures and scenes. This makes the device instantly accessible to young children and people with mental disabilities who may be unable to grasp written language.

Talkitt. This application helps people with speech and language disorders to communicate with clarity. It translates unintelligible pronunciation into understandable speech, so the other person can understand what the user said. It works by learning the user's speech patterns first, creating a personal speech dictionary. Then it identifies and recognises the unintelligible pronunciations, translating them into speech people can understand.

Technologies for others

AXS map. Many public places are not equipped with facilities like wheelchair ramps and wheelchair-accessible restrooms. This causes inconvenience to those who need a wheelchair

to move around. AXS Map is a crowd-sourced map that provides information about wheelchair-accessible ramps and restrooms in public places such as restaurants, hotels and shopping malls. Star ratings help users know how well-designed these facilities are.

Assist-Mi. This app connects service providers and caregivers with specially abled users on the go. Users can get assistance in real time at the touch of a button. Services include help in getting to work, shopping and travelling. The Mi-Profile feature of the app lists users' needs so service providers know what to do when assistance is requested. The app also supports two-way communication and GPS for better information on location.

Lifeware. It is a self-stabilising handle that can be attached to eating utensils like forks and spoons. It is very helpful for people with Parkinson's disease or other forms of motion disorders that cause hand tremors. Lifeware stabilises up to 70 per cent of the disruption and helps reduce the amount of spillage before food reaches the patient's mouth. The Lifeware system comes with a charger and three utensils, a spoon, a fork and a soup spoon in addition to the handle. Depending on how long the user's meals last, Lifeware lasts for three to six meals between charges. The handle can be wiped clean, while the spoons and fork can be washed like a normal utensil.

HeadMouse Nano. HeadMouse Nano replaces the standard computer mouse for people who cannot use or have limited use of their hands. Mounted on the top of a laptop's screen and showing an onscreen keyboard, it tracks and translates natural movements of the user's head into directly proportional mouse pointer movements—move your head and the mouse pointer moves as well. The HeadMouse has a wireless optical sensor that tracks a tiny disposable target worn